Aerospace Science and Technology 72 (2018) 305–315

Contents lists available at ScienceDirect

Aerospace Science and Technology

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An autonomous aerial combat framework for two-on-two engagements based on basic fighter maneuvers

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ARTICLE INFO

Article history:
Received 15 June 2017
Received in revised form 7 November 2017
Accepted 7 November 2017
Available online 13 November 2017

Keywords:
Aerial combat
Unmanned combat aerial vehicle
Two-on-two engagement
Basic fighter maneuver
Cooperative tactic

ABSTRACT

Unmanned combat aerial vehicles are expected to assist human pilots or to perform autonomous aerial combat in the future. Previous research studies on autonomous aerial combat often have limitations due to overly simplifying assumptions, heavy computational loads, and limited scalability. In this paper, we propose a novel autonomous aerial combat algorithm with high-performance and real-time calculations based on basic fighter maneuvers for gun-based engagement within visual range. From the perspective of energy maneuver, which is the core concept of basic fighter maneuvers, we analyzed the flight envelope and maximum maneuvering point of a fighter jet model using the velocity-load factor and energy-maneuver diagrams. We designed a score function and matrix for situation assessment based on combat geometries, enabling cooperative maneuvers and role assignment. By performing gun modeling and target prediction based on the maneuvering plane, the impact point for gun shooting was obtained. The performance of the designed algorithm was validated through not only Monte-Carlo simulations with various initial conditions, but also engagements with another algorithm for comparison, and evaluation sessions by an Air Force combat instructor in a real-time simulation environment using virtual reality goggles and X-Plane.

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1. Introduction

Next-generation unmanned combat aerial vehicles (UCAVs) are considered promising indispensable assets in future warfare, eliminating potential threats to the lives of pilots through assistance in or performance of autonomous combat. In this study, an autonomous aerial combat framework for UCAVs is designed to mimic the human pilot for effective decision-making and tactical maneuver generation.

This study focuses on gun-based aerial combat within visual range (WVR), referred to as *dogfighting*. Although missiles have been developed as the primary equipment for beyond-visual-range combat, especially for second-generation fighters and later [1], the efficiency and number of fatalities by missiles are below expectations [2]. Thus, as next-generation fighters, UCAVs with guns are considered essential for WVR combat. Particularly, gun-based combats are highly-constraining problem requiring more precise and agile maneuverers compared to missile-based engagement. In addition, effective one-to-one maneuvering does not ensure tactical suitability in two-on-two environments. Unlike one-on-one dogfight, which simply aims to capture the opponent's tail, the complexity of two-on-two engagements is exponentially greater. The algorithm proposed in this study generates cooperative tactical maneuvers based on a score function, target position prediction, and target/role assignment fitting for such complex environment.

Various studies on autonomous aerial combat have been performed since early 1960s. Combat problems are modeled as simple pursuit–evasion games, and various theoretical approaches and optimal control schemes provide solutions for autonomous aerial combat. Aerial combat was modeled as a deterministic, perfect-information pursuit–evasion game using differential game theory [3–5]. A general pursuer-evader game model was solved using dynamic programming, which led to the principle of optimality [6–10]. An indoor flight test limited to level flying was also performed with optimal policy using approximate dynamic programming [11]. A more complex two-on-one combat was treated as a differential game in three-dimensional space [12]. Nonlinear control and guidance laws, including model predictive control [13, 14] and sliding-mode guidance [15], were proposed to determine the solutions for uncertain combat situations. These theoretical approaches enable the mathematical interpretation of aerial combat

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https://doi.org/10.1016/j.ast.2017.11.014
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problems. However, they may include oversimplifying assumptions, such as fixed roles or two-dimensional motion, to alleviate the complexity and computation time.

Other approaches for modeling autonomous combat are heuristic rule-based systems that imitate the behavior of human pilots. The responsive maneuvers for each combat situation were regularized, and the approaches were validated through combat simulations with human pilots. NASA proposed the rule-based adaptive maneuvering logic, called Paladin, which showed a competitive aerial combat performance with human pilots [16–20]. In the same way, an influence diagram game was used to model the decision process of a pilot during air combat to choose the right maneuvers at each moment [21]. The rule-based system produced realistic and sensible simulation results. However, dealing with rules for all combat situations, modifying previously designed rules, and adding new rules are difficult.

Reinforcement learning and genetic algorithms with classifier systems have recently exhibited an improved performance by generating effective new tactics for various simulation environments [22–24]. Using genetic based machine learning, these algorithms demonstrated the ability to discover model-free tactics in an air combat environment [22,23]. The Q-learning method generates novel actions with a designed action value function and enables a successful outmaneuvering of the target [24]. However, enormous number of training combat cases are required to attain a competitive level of performance. In addition, the rules and maneuvers generated not only make it challenging to verify their physical meaning and optimality, but are also limited when attempting to ensure a consistent performance.

Further practical improvements are required for gun-based WVR autonomous aerial combat. This paper proposes a novel solution that enables real-time and high-performance autonomous combat in a complex two-on-two environment. Combat performance is ensured by introducing a basic fighter maneuver (BFM) concept that suggests ideal maneuvers for various combat situations based on information accumulated by pilots [25,26]. Specifically, a modern energy management technique is developed based on an energy–maneuver (E–M) chart and a velocity–load factor (V–n) diagram. In this technique, situation assessment is performed using the designed score function, enabling cooperative tactical maneuvers in complex two-on-two engagement. For a preemptive response to an enemy, the future target position is predicted based on the maneuvering plane, and the accurate impact point for striking with a gun is generated based on gun modeling. The proposed autonomous combat algorithm is validated through high fidelity model based Monte-Carlo simulations with various initial conditions. In addition, a virtual engagement with a comparative algorithm and an evaluation by an Air Force combat instructor in active duty through virtual reality (VR) goggles and an X-Plane-based real-time simulation environment are performed.

The remainder of this paper is organized as follows. Section 2 explains and defines the combat geometry and its parameters. Next, the specific theory and techniques for autonomous aerial combat are described in Section 3. Section 4 details the virtual combat environment and analyzes the performance of proposed algorithm. This paper is concluded in Section 5. In this study, the UCAV piloted by the proposed algorithm and the opponent aircraft are referred to as the *attacker* and the *target*, respectively, for simplicity.

2. Gun-based aerial combat problem statement

In gun-based combats, the basic objectives are to position itself at a tactically advantageous location to attack the target while avoiding being shot by other hostile fighter jets. Such attack points change continuously in three-dimensional space according to the maneuvering of the attacker and target. In order to describe these changes, the combat geometry and parameters shown in Fig. 1 are employed.

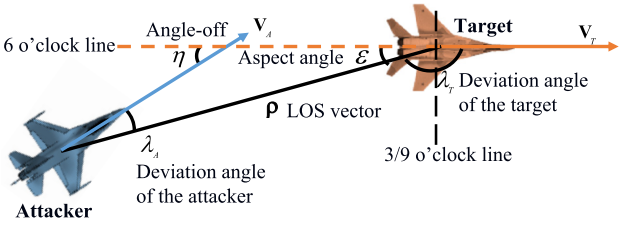


Fig. 1. Combat geometry and parameters.

The angle-off η represents the extent of the distortion of the fuselages of the attacker and target and can be obtained in terms of the velocities of the attacker $\mathbf{V}_A$, and target $\mathbf{V}_T$, assuming that side slip is negligible. The aspect angle ε indicates the distance of the attacker from the 6 o'clock line of the target [27], as depicted in Fig. 1. The deviation angle λ_A represents the angular difference between the heading of the fighter and $\boldsymbol{\rho}$. These three parameters can be obtained from Eq. (1), where $\boldsymbol{\rho}$ denotes the line-of-sight (LOS) vector from the attacker to the target:

$$\eta = \psi_A - \psi_T = \cos^{-1}\left[\frac{\mathbf{V}_A \cdot \mathbf{V}_T}{|\mathbf{V}_A||\mathbf{V}_T|}\right], \quad \varepsilon = \cos^{-1}\left[\frac{\mathbf{V}_T \cdot \boldsymbol{\rho}}{|\mathbf{V}_T||\boldsymbol{\rho}|}\right],$$
$$\lambda_A = \cos^{-1}\left[\frac{\mathbf{V}_A \cdot \boldsymbol{\rho}}{|\mathbf{V}_A||\boldsymbol{\rho}|}\right] \tag{1}$$

3. Design of the modular components of the algorithm

The autonomous combat algorithm proposed in this study performs the roles of a human pilot in a fighter jet and conducts situation assessment, role assignments, and combat maneuver generation. In this section, each modularized component constituting the algorithm is described, along with its simulation environment. The algorithm proposed in this paper uses only the observable velocity vector and relative distance of the target, which real fighter jets provide at a rate of 50 Hz [28], to ensure practicality for real applications.

3.1. Situation assessment

The first task of the pilot when engaging in aerial combat is to assess the situation. BFM-based situation assessment classifies combat situations into four categories [27]. This technique is discontinuous because it cannot describe situations other than those in the predefined categories, and it excludes qualitative information. This classification is not suitable for the algorithm; thus, a score function is proposed to quantify combat situations.

From Fig. 1, it can be seen that a smaller value of $|\lambda_A|$ implies that the attacker or the gun is aimed more accurately at the target, while $\pi - |\lambda_A|$ is proportional to the chance of a shot or an offensive advantage. Similarly, a smaller value of $|\lambda_T|$ indicates a higher probability of being shot by the enemy; thus, $|\lambda_T|$ is proportional to the survival advantage. Based on these features, the following cost function in Eq. (2) is designed to quantify a combat situation in terms of the deviation angle:

$$0 \leq \text{Score} = \frac{k(\pi - |\lambda_A|)^2 + (1-k)|\lambda_T|^2}{\pi^2} \leq 1 \tag{2}$$

Different pilots may assess a given situation differently based on their own experience and intuition. Hence, eq. (2) includes the parameter k, which is the weighting gain representing aggressiveness. Figure 2 presents the score surfaces and extreme cases evaluated by situation assessment with respect to k.

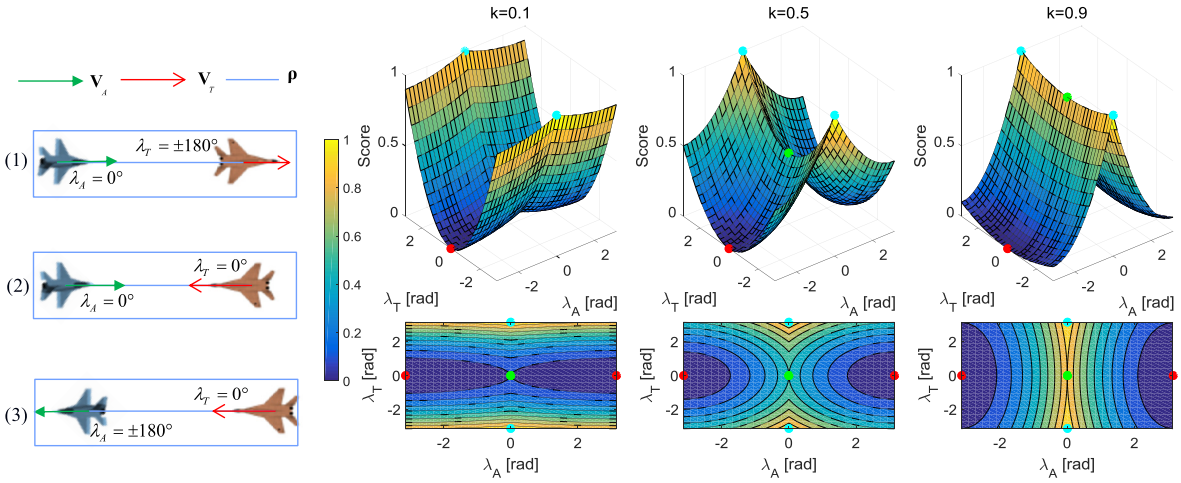


Fig. 2. Score surfaces and extreme cases assessment based on three different k values.

The designed score function yields continuous values and surfaces reflecting the qualitative depth for all combat geometries. The score surface when k has a small value of 0.1 mainly determines the situation according to λ_T. However, the score value changes more relative to λ_A as k increases because the score function of the situation assessment has different weights between offensive and survival advantages depending on the value of k. Regardless of the value of k, the extreme cases of $[0,\ \pm\pi]$ and $[\pm\pi,\ 0]$, which are depicted as (1) and (3) in Fig. 2, are evaluated as cyan and red points in the surfaces, and as ideal points of score 1 and worst points, respectively. However, the head-on case described as (2) in Fig. 2 could be assessed differently depending on the k value as the green point in the surfaces. We set a k as 0.5 herein for evaluating the attack and defensive maneuvers evenly without being shifted to one side.

3.2. Cooperative tactics for two-on-two engagements

The solution for one-on-one engagements cannot be considered a sufficient condition for many-to-many combat problems, which do not simply consist of multiple pairs of one-on-one combat situations. Therefore, cooperative tactics are necessary for improving the survival rate and defeating the targets quickly. To accomplish these objectives, a score matrix whose components are derived from Eq. (2) can be constructed to determine the survival/offensive advantages of the attacker squadron. Equation (3) shows the score matrix for two-on-two engagement. Each component of the score matrix represents the score of the attacker in a row against the target in a column obtained from Eq. (2). For example, S_{ij} denotes the score of attacker A_i against target T_j:

$$\text{Score matrix} \triangleq \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \tag{3}$$

In this study, each fighter on the attacker side is allowed to select the object of attack and to determine its own role actively during the cooperative stratagem based on the score matrix. At this point, priority is given to eliminating the target with a higher score and outnumbering the enemy. This strategy is consistent with the tactics of human pilots: outnumber the enemy in the air first, then outmaneuver to gain firing position.

When both attackers are in superior position, each attacker performs offensive maneuvers against the easily winnable target. However, if at least one of the two attackers has the possibility of being shot at with a low score, cooperative tactics should be carried out to increase the survivability of friendly forces and to eliminate the target quickly. A tactical maneuver is performed by allocating the roles of a free fighter and a bait to eliminate the target. To do this, a target index J_A with a low degree of threat and ease of being shot down is found that maximizes the squared score sum of the columns, $\sum_i S_{ij}^2$, for each target T_j, as shown in Eq. (4):

$$J_A = \arg\max_j \sum_i S_{ij}^2 \tag{4}$$

To perform tactical combat on the selected target, the attackers are assigned the roles of either a bait or a free fighter, similar to the tactics of human pilots [29]. While one attacker acting as bait engages the target, the other attacker, acting as a free fighter with a high survival rate and the possibility of striking, performs an offensive maneuver to attempt to eliminate the target quickly. The indices of the bait I_B and the free fighter I_{FF} relative to the assigned target J_A can be obtained through Eq. (5):

$$I_B = \arg\min_i S_{iJ_A}, \qquad I_{FF} = \arg\max_i S_{iJ_A} \tag{5}$$

The assigned free fighter and bait are expected to perform offensive and defensive maneuvers according to the combat guidance law described in Section 3.4.

3.3. Target position and impact point prediction

A reasonable prediction for the future position of the target is essential for preemptive actions to target maneuvers. However, an accurate prediction is difficult because of the uncertainty of the target model and intention. The exact model and features of the combat aircraft are completely concealed, and the combat strategy has unpredictable randomness because of the individual tendency of each pilot. Erroneous predictions can worsen a combat situation; hence, target prediction should be carefully performed. To cope with these external uncertainties, the target is assumed to perform reasonable and general decisions and maneuvers based on the BFM concept. To achieve these objectives, the purpose of the maneuvers of an opponent is analyzed for a given engagement situation and applied to the maneuvering plane to enable reasonable prediction of the future position of the target with the following two assumptions: 1) the target performs reasonable and general decisions and maneuvers based on the proposed BFM-based algorithm; 2) the predicted maneuver is maintained for the look-ahead time T_{LAH}.

3.3.1. Estimation of T_{LAH}

The size of the prediction interval directly affects the accuracy and computational complexity of the target prediction. In this

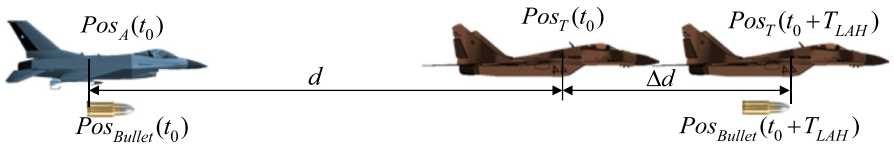
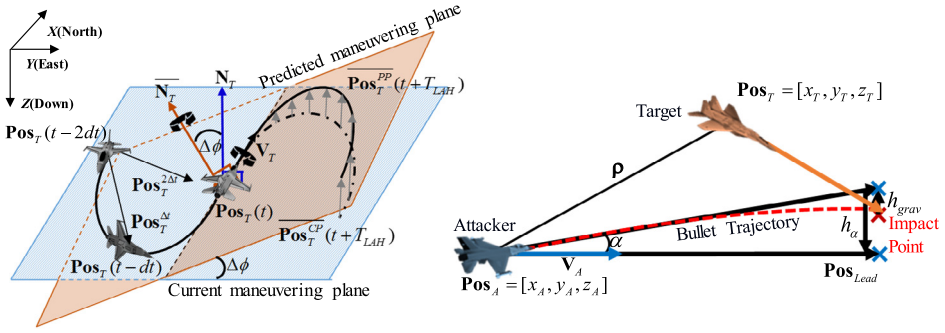


Fig. 3. Relationship between the bullet launched from the attacker and the maneuvering target.



a) Target prediction based on the maneuvering plane b) Geometry of the impact and the lead pursuit point

Fig. 4. Target position prediction and lead pursuit point.

study, we set the time of flight (TOF) required for a bullet to reach the enemy after the gun is fired as T_{LAH} based on Fig. 3.

If the attacker fires a gun toward the target at t_0, the TOF until the bullet hits the target can be obtained by summing the times to fly the initial distance d and Δd by which the target has moved beyond d after the shot is fired, as shown in Eq. (6). T_{LAH} is expressed as a function of muzzle velocity V_{Muzzle}, which is set to 1030 m/s based on [30], and the target speed V_T:

$$V_{Muzzle}T_{LAH} = d + \Delta d = d + V_T T_{LAH}, \quad T_{LAH} = \frac{d}{V_{Muzzle} - V_T} \tag{6}$$

3.3.2. Target prediction and impact point

In this study, the maneuvering plane is used to predict the future target position. To perform this prediction, we define two vectors from the past flight trajectory, and the normal vector $\mathbf{N}_T$ for the current maneuvering plane can be obtained as in Eq. (7) and Fig. 4a):

$$\begin{aligned}
&\mathbf{Pos}_T^{2\Delta t} = \mathbf{Pos}_T(t) - \mathbf{Pos}_T(t-2dt),\\
&\mathbf{Pos}_T^{\Delta t} = \mathbf{Pos}_T(t-dt) - \mathbf{Pos}_T(t-2dt)\\
&\mathbf{N}_T = \mathbf{Pos}_T^{2\Delta t} \times \mathbf{Pos}_T^{\Delta t}
\end{aligned} \tag{7}$$

The BFM-based algorithm proposed in this study is applied to the target in reverse to predict the intention of the target maneuvers, $\overline{N_{Zcmd}}$ and $\overline{p_{cmd}}$, by assumption 1) above. When the target performs the interpreted maneuvers during T_{LAH}, as in assumption 2), a new $\overline{\mathbf{N}_T}$ for the rotated new maneuvering plane can be obtained by rotating $\mathbf{N}_T$ by $\overline{p_{cmd}}T_{LAH}$ about the axis of rotation $\mathbf{V}_T$ with the Rodrigues rotation formula:

$$\begin{aligned}
\mathbf{r} = \frac{\mathbf{V}_T}{|\mathbf{V}_T|}, \qquad \overline{\mathbf{N}_T} &= \mathbf{N}_T\cos(\overline{p_{cmd}}T_{LAH}) + (\mathbf{r}\times\mathbf{N}_T)\sin(\overline{p_{cmd}}T_{LAH})\\
&+ \mathbf{r}(\mathbf{k}\cdot\mathbf{N}_T)\big(1-\cos(\overline{p_{cmd}}T_{LAH})\big)
\end{aligned} \tag{8}$$

In the same way, the estimated velocity vector $\overline{\mathbf{V}_T}$ reflecting the maneuver intention $\overline{N_{Zcmd}}$ can be obtained by rotating the current velocity vector $\mathbf{V}_T$ by $\overline{N_{Zcmd}}T_{LAH}$ about the axis of rotation $\overline{\mathbf{N}_T}$, as in Eq. (9):

$$\begin{aligned}
\mathbf{r} = \frac{\overline{\mathbf{N}_T}}{|\overline{\mathbf{N}_T}|}, \qquad \overline{\mathbf{V}_T} &= \mathbf{V}_T\cos(\overline{N_{z_{cmd}}}T_{LAH}) + (\mathbf{r}\times\mathbf{V}_T)\sin(\overline{N_{z_{cmd}}}T_{LAH})\\
&+ \mathbf{r}(\mathbf{k}\cdot\mathbf{V}_T)\big(1-\cos(\overline{N_{z_{cmd}}}T_{LAH})\big)
\end{aligned} \tag{9}$$

Assuming that $\overline{\mathbf{V}_T}$ placed on the new maneuvering plane lasts for T_{LAH}, the future target position $\overline{\mathbf{Pos}_T^{PP}}$ can be reasonably estimated relative to $\overline{\mathbf{Pos}_T^{CP}}$, as shown in Eq. (10) and Fig. 4a):

$$\overline{\mathbf{Pos}_T^{PP}}(t+T_{LAH}) = \mathbf{Pos}_T(t) + \overline{\mathbf{V}_T}T_{LAH} \tag{10}$$

A feasible lead pursuit point $\mathbf{Pos}_{Lead}$ can be obtained for accurately aiming at the impact point, which is promising position for striking the target, by using Eq. (11) and incorporating the alteration in the trajectory due to gravity drop h_g, the distance h_α caused by angle of attack α, and $\overline{\mathbf{Pos}_T^{PP}}$. The geometry considering gun model, impact point, and lead pursuit point is described in Fig. 4b).

$$\begin{aligned}
\mathbf{Pos}_{Lead} &= \overline{\mathbf{Pos}_T^{PP}}(t+T_{LAH}) - h_{grav} + h_\alpha\\
&= \overline{\mathbf{Pos}_T^{PP}}(t+T_{LAH}) - \left[0, 0, \frac{1}{2}gT_{LAH}^2\right]^T\\
&+ C_b^r(\phi_A, 0, 0)[0, 0, d\tan\alpha]^T
\end{aligned} \tag{11}$$

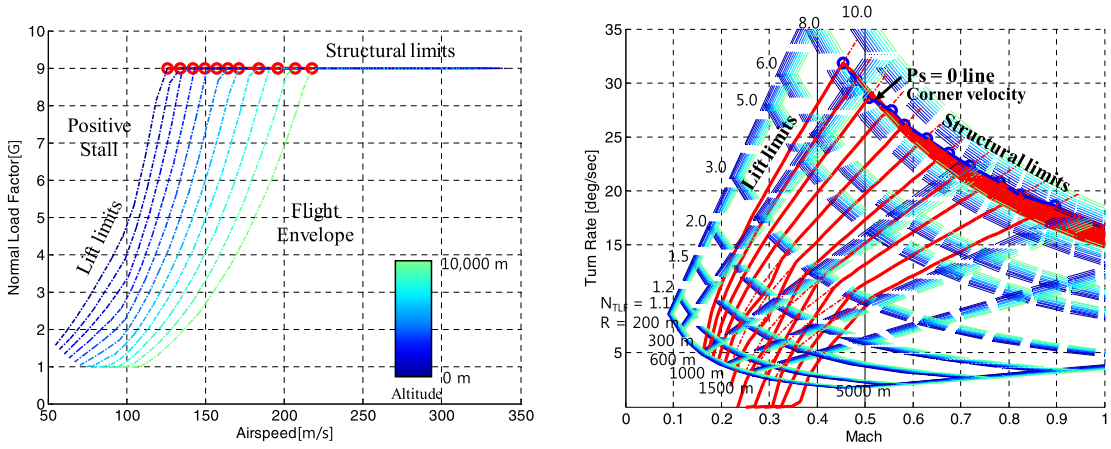
3.4. Model-based autonomous aerial combat guidance

3.4.1. Combat aircraft model analysis

In this study, flight envelope analysis and interpretations of the corner velocity with the maximum turning rate are performed using the following two methods: 1) utilizing a V–n diagram to analyze the turn-performance capabilities in nominal velocity regions and flight envelopes, and 2) employing an E–M chart to determine the maximum turn rate, radius, and velocity based on the flight conditions. Figure 5 shows the V–n diagram and E–M chart for an F-16 model [31] at an altitude of 10,000 meters in 1,000 meters intervals.

The V–n diagram explains the relationship between the load factor and airspeed and is among the most common and effective technique for investigating the performance of a model. The contact points of the structural strength limits and lift limits can be defined as corner velocities. The E–M theory explains energy management for WVR aerial combat and ensures the high-performance of the BFM [26]. The turn radius R and turn rate TR can be obtained using Eq. (12):

$$R = \frac{V^2}{g\tan\phi} = \frac{V^2}{g\sqrt{n^2-1}}, \qquad TR = \frac{g\sqrt{n^2-1}}{V} \tag{12}$$



a) V–n diagram and flight envelope b) E–M chart and corner velocity

Fig. 5. Fighter jet model analysis based on an F-16 database.

The maximum turn rate $TR_{\max}$ is achieved when $C_L = C_{L\max}$ and $n = n_{\max}$. The specific energy E_s and specific excess power P_s, formulated in Eq. (13), are used to explain maneuverability considering the energy and its rate of change:

$$E_S = \frac{E}{mg} = h + \frac{V^2}{2g}, \quad P_S = \frac{dE_S}{dt} = \frac{dh}{dt} + \frac{V}{g}\frac{dV}{dt} = \frac{T-D}{mg}V \tag{13}$$

Based on Eq. (13), an energy maneuver can be evaluated for a given weight, thrust, velocity, altitude, and load factor, and the corner velocity V_C can be obtained by considering $P_S = 0$. This is the point of intersection for the maximum load factor and the stall limit lines and implies that it has the maximum turn rate and minimum turn radius, as shown in Fig. 5.

3.4.2. Aerial combat guidance law

The autonomous combat guidance law uses the analyzed model characteristics to generate offensive and defensive maneuvers based on the BFM. In the offensive geometry shown in Fig. 6a), if an attacker tracks only $\mathbf{Pos}_D$, which is the current desired position relative to the target performing evasive maneuvers along the solid red line, significant delays can occur, and an advantageous situation can become neutral or even disadvantageous, as observed by the dotted blue line. Therefore, this study proposes a combat guidance law to perform BFM considering the turn circle and entry window [27]. The position command $\mathbf{Pos}_{cmd}$ for passing through an entry window and achieving a gun-shot condition by following blue solid line is obtained based on $\mathbf{Pos}_{Lead}$, the proper gun range d_{gun}, and the compensation value of the error $\mathbf{e}_p$ from the current ideal position multiplied by the gain K_e as in Eq. (14) and Fig. 6:

$$\begin{aligned}&\mathbf{Pos}_D = \mathbf{Pos}_T + \mathbf{C}_b^r(\phi,\gamma,\chi)[d_{gun}\ 0\ 0]^T, \qquad \mathbf{e}_P = \mathbf{Pos}_D - \mathbf{Pos}_A\\ &\mathbf{Pos}_{cmd} = \mathbf{Pos}_{Lead} + \mathbf{C}_b^r(\phi,\gamma,\chi)\left(K_e\mathbf{e}_p + [\mathrm{d}_{gun}\ 0\ 0]^T\right)\end{aligned} \tag{14}$$

The velocity command $\mathbf{V}_{cmd}$ required by the attacker to follow and attain an offensively advantageous position $\mathbf{Pos}_{cmd}$ can be obtained based on the proportional and differential gains, K_{VP} and K_{VD}, as in Eq. (15):

$$\mathbf{e}_V = \mathbf{V}_T - \mathbf{V}_A, \quad \mathbf{V}_{cmd} = \mathbf{V}_T + \mathbf{C}_b^r(\phi,\gamma,\chi)(K_{VP}\mathbf{e}_P + K_{VD}\mathbf{e}_V) \tag{15}$$

In addition, $\mathbf{a}_{cmd}$ can be obtained using Eq. (16) to follow $\mathbf{Pos}_{cmd}$:

$$\boldsymbol{\omega} = \frac{1}{|\mathbf{Pos}_{cmd} - \mathbf{Pos}_A|}\mathbf{C}_b^r(\phi,\gamma,\chi)\frac{\boldsymbol{\rho}}{|\boldsymbol{\rho}|} \times \mathbf{e}_V,$$

$$\boldsymbol{\rho}_{cmd} = \mathbf{Pos}_{cmd} - \mathbf{Pos}_A$$

$$\mathbf{a}_{cmd} = K_{PG}\frac{\mathbf{V}_A \times \boldsymbol{\rho}_{cmd} \times \mathbf{V}_A}{\mathbf{V}_A|\boldsymbol{\rho}_{cmd}|g} + K_{PNG}\frac{\mathbf{e}_V[0\ \omega(3) - \omega(2)]^T}{g} \tag{16}$$

The acceleration command in Eq. (16) can be obtained by introducing an LOS-vector-based guidance law. The rate of change of the LOS is compensated by incorporating a proportional navigation guidance term to cope with the maneuvers of the target. This combined guidance law also exhibits robustness against noise and the maneuvers of the target [32].

As aerial combat occurs in fast-flight regions, assuming no side slip and a horizontal maneuvering plane, p_{cmd} and Nz_{cmd} can be obtained using Eq. (17):

$$Nz_{cmd} = [-1\ 0\ 0]\mathbf{a}_{cmd}, \quad p_{cmd} = \tan^{-1}\frac{[0\ 1\ 0]\mathbf{a}_{cmd}}{Nz_{cmd}} \tag{17}$$

A BFM performed by the algorithm utilizes the model properties, including V_C and $TR_{\max}$, more actively when a defensive maneuver is performed in a bait role through the score function and role assignment. This is because evasion must be performed through the maximum maneuvering point to escape the area in which an enemy can attack. The guidance command for defense is obtained using Eq. (18):

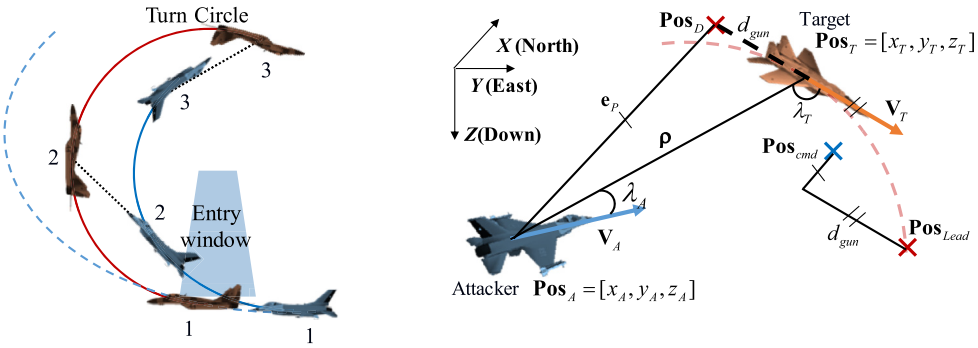
$$R_{\min} = \frac{\mathbf{V}_A}{TR_{\max}}, \quad \phi_{cmd} = sign(\rho)\tan^{-1}\frac{\mathbf{V}_A}{R_{\min}g}, \quad V_{cmd} = V_C \tag{18}$$

The algorithm proposed in this study enables flight control to follow the velocity command without stalling through the energy management technique of altitude–velocity conversion. Assuming that the energy is converted only into kinetic and potential energy for the fighter model treated as a point mass, the required altitude displacement Δh can be obtained using the altitude, velocity, and total energy E by applying Eq. (19):

$$\begin{aligned}&E = mgh + \frac{mV^2}{2}, \quad E' = mgh' + \frac{mV_{cmd}^2}{2},\\ &\Delta h = h - h' = \frac{V^2 - V_{cmd}^2}{2g}\end{aligned} \tag{19}$$

In addition, the kinetic energy induced by the rate of climb for the attacker and target during combat must be considered using Eq. (20):

$$\Delta h_r = \frac{(V_A\sin\gamma_A)^2}{2g}sign(\sin\gamma_A) - \frac{(V_T\sin\gamma_T)^2}{2g}sign(\sin\gamma_T) \tag{20}$$



a) Passing entry window during offense b) Offensive combat geometry

Fig. 6. Offensive maneuver based on BFM using the predicted target position.

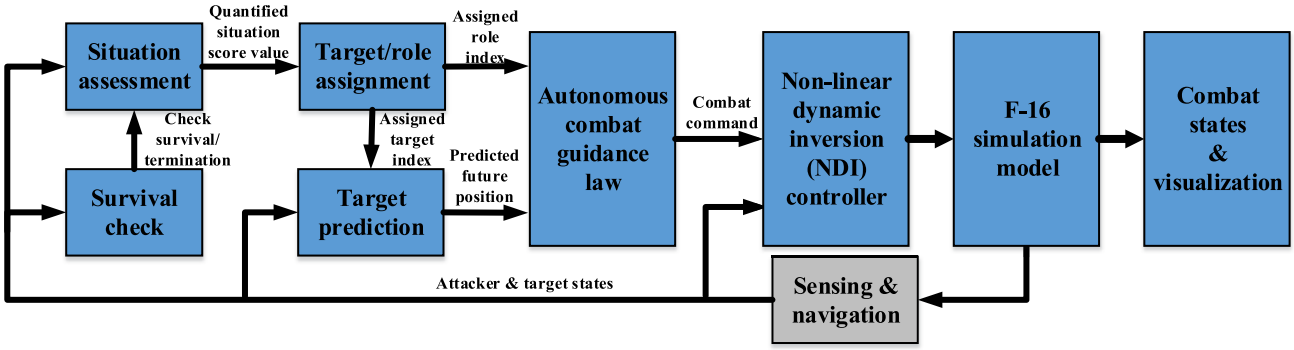


Fig. 7. Schematic of the designed algorithm.

As a result, the value of altitude compensation that can accurately follow the velocity command through energy management without falling into a stall during high-alpha elevation can be obtained using Eq. (21):

$$h'_{cmd} = h_{cmd} + (\Delta h + \Delta h_r) \tag{21}$$

4. Aerial combat simulations

To validate the autonomous combat performance, scalable combat simulation environments were developed based on the description provided in Section 3 with the modular structure shown in Fig. 7. In this study, an aerodynamic model and thrust data for the F-16 from [31] are used, and the translational dynamics along the y-axis are ignored by assuming the side slip angle to be zero ($\beta = 0$). This 5-DoF model not only has sufficiently high fidelity to replace a 6-DoF full dynamics model [33], but also is more suitable for use in real-time simulation environments.

In addition, a non-linear dynamic inversion (NDI) controller is used as a baseline controller and steers the system states to precisely track the generated combat maneuvers. The NDI controller removes the nonlinearities from the model by directly inverting the original dynamics, such that it works as a self-scheduled controller and non-parametric architecture [34,35]. Therefore, the NDI controller ensures stability and adequate performance in aerial engagement, which occurs in a wide range of operational regions and can cope with internal uncertainties. The model is updated using the Runge–Kutta fourth-order integration equation to avoid the singularity problem of the Euler angle and conducted for one hundred seconds at one hundred Hz.

4.1. Combat simulation scenarios

The first scenario sets a one-on-one tracking situation to ensure high yo-yo energy managing maneuvers to prevent overshoot. In addition, complicated two-on-one and two-on-two virtual combats are planned to validate the performance of the cooperative maneuvers. We assign a tail-chased conditions, and the attackers attempt to reverse the adverse initial conditions and outmaneuver the enemies through cooperative tactical maneuvering. The initial conditions for each representative combat scenario and the associated geometric relationships are presented in Table 1 and Fig. 8.

Furthermore, Monte-Carlo simulations were conducted with one thousand times of virtual combats, having uniformly distributed initial score values and distances, to guarantee the universal validity of the performance of the proposed algorithm. For the purpose of explanation, the attackers controlled by the designed algorithm are indicated in blue and cyan, and the targets are indicated by red and magenta.

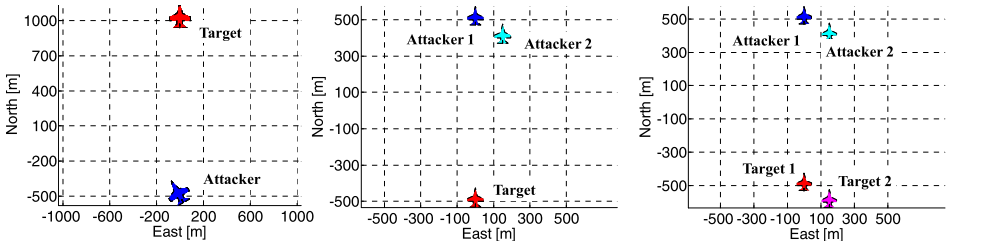
4.2. Simulation results and analysis

The results of the first simulation scenario are presented in Fig. 9. The plane above the trajectory is depicted at intervals of five seconds. To reduce the initial relative distance of 1,500 m, the attacker gained its velocity by thrusting and decreasing its altitude for 35 seconds. After that, the combat algorithm raised the nose and increased the altitude for energy exchange while lowering the velocity; thus, the attacker had the same velocity as the target in 60 seconds. The described high yo-yo, which prevented overshoot, can clearly be seen in Fig. 9b). As a result, the target was successfully destroyed in 80 seconds by gun shot.

The second combat scenario consists of an outnumbered attacker team traced by a target. Since the attacker side did not know the intentions of the enemy, they employed a defensive split, as shown in Fig. 10a), which is an ideal defensive maneuver commonly used by human pilots [29], along with a corner velocity and maximum turn rate. Based on the situation assessment and score value, attacker 1 and 2 were assigned with the roles of bait and a free fighter, respectively, as shown in Fig. 10c). As a result of

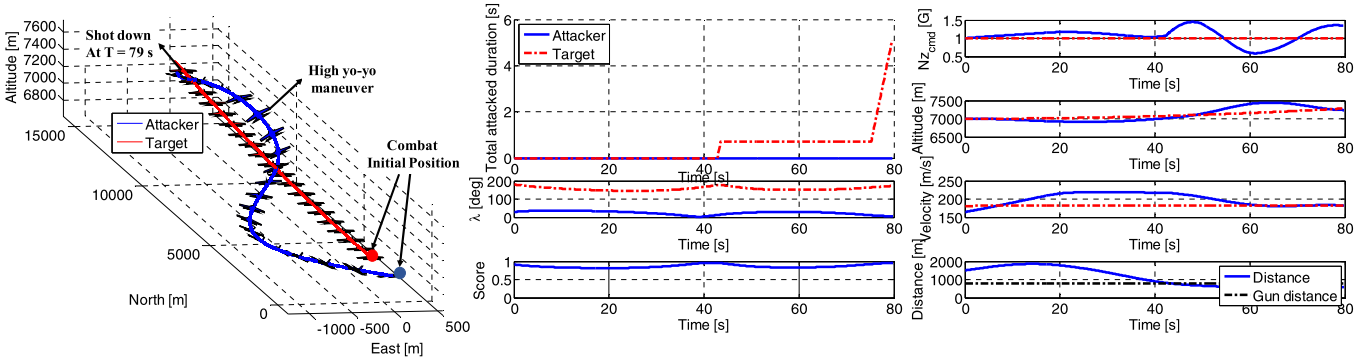
Table 1
Representative combat simulation scenarios and initial conditions.

Simulations scenarios	Description of situation	Initial combat conditions			
		(V_A), (V_T) [m/s]	(h_A), (h_T) [Kms]	Distance [m]	Target maneuver
#1. 1:1	Energy managing high yo-yo	(165), (180)	(7), (7)	1,500	Level flight
#2. 2:1	Strategic maneuvering	(250, 250), (270)	(7, 7), (8)	1,353	Pure pursuit guidance
#3. 2:2		(250, 250), (270, 270)	(7, 7), (8, 8)	1,353	



a) One-on-one initial positions b) Two-on-one initial positions c) Two-on-two initial positions

Fig. 8. Initial geometry conditions during representative combat simulations.



a) High yo-yo trajectories b) High yo-yo combat result plots

Fig. 9. Scenario 1 – One-on-one high yo-yo simulation results.

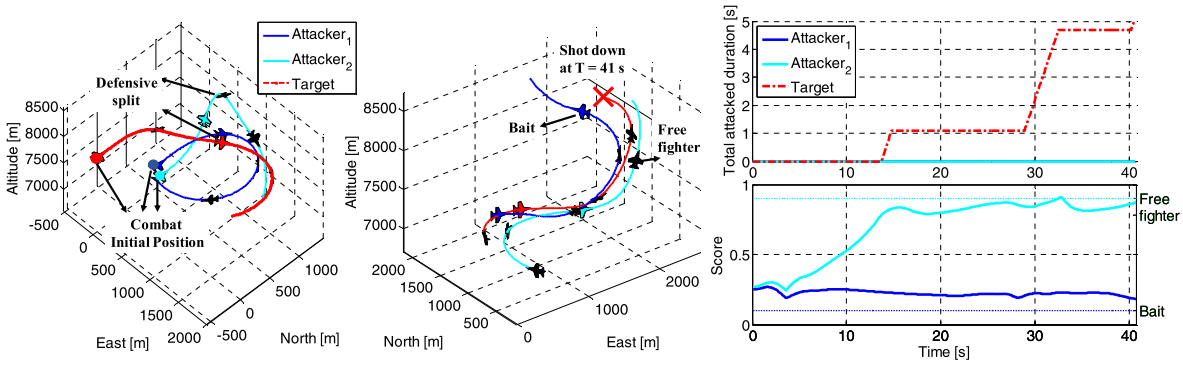
this tactical maneuver, the attacker team achieved favorable conditions with a score value exceeding 0.8 in 15 seconds. When the bait escaped from the gunnery area of the enemy, the free fighter traversed the entry window based on the target prediction information and performed a successful attack, as shown in Fig. 10b).

In addition, one thousand Monte-Carlo simulations with a uniformly distributed initial score values and distances were performed to ensure the universal validity of the performance in two-on-one combats. When the score was less than 0.1 in the initial conditions, these unfavorable conditions could be reversed to neutral or victory in 79 percent out of 117 cases. In the remaining 883 cases with initial scores of 0.1 or greater, the attacker team performed ideal cooperative maneuvers that successfully eliminated the opponent without any single loss of an attacker.

Finally, we validated the performance of the designed algorithm in two-on-two combat, with unfavorable initial tail-chased condition. Since the intentions of the enemies were unknown, the attacker team started the tactical strategy by performing a defensive split to identify the plan and intention of the bandit squad, as shown in Fig. 12a). After a transition period of 8 to 20 seconds, target 2 was selected as the object of an attack based on the total score value at 20 seconds, as shown in Fig. 12c). Simultaneously, the attacker team assigned roles in such a way that attacker 1, with a score higher than 0.8, acted as the free fighter, and the other, with a score of approximately 0.3, acts as bait, as shown in Fig. 12c). Finally, the free fighter passed the entry window by using the maneuvers generated based on BFM and shots target 2 in 99 seconds.

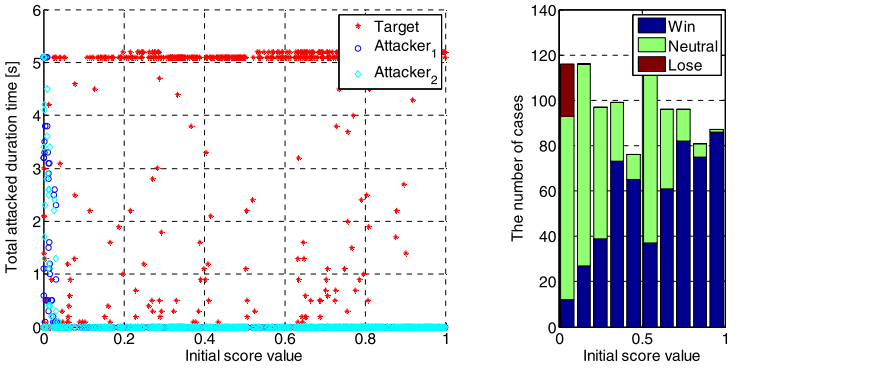
To validate the universal performance of the proposed algorithm in complicated two-on-two combat, one thousand Monte-Carlo simulations with randomly uniformly distributed initial scores and distances were conducted, and the results are presented in Fig. 13.

The two-on-two engagement results are much more complicated than those of the two-on-one engagements in Fig. 11. It can be seen in Fig. 13a) that shooting cases occurred more frequently. The frequency of being shot by the targets decreased sharply as the score became 0.2 or higher, and the given adverse circumstances were turned into neutral or favorable situations. This tendency can be seen more clearly in Fig. 13b). For scores of less than 0.2, the probability of losing at least one attacker is more than 90%. However, the proposed algorithm performed proper cooperative maneuvers, causing the probability of winning or neutralizing to be greater than 80 percent with a score value exceeding 0.2. It was also confirmed that the probability of defeating the target squadron was more than 80 percent for a score of 0.7 or more, which is superior to the non-cooperative pursuit guidance law of the comparative target squadron. These last simulations demonstrate that the proposed algorithm can advantageously reverse combat situations with universal validity despite unfavorable initial conditions, in complicated two-on-two combat environments.



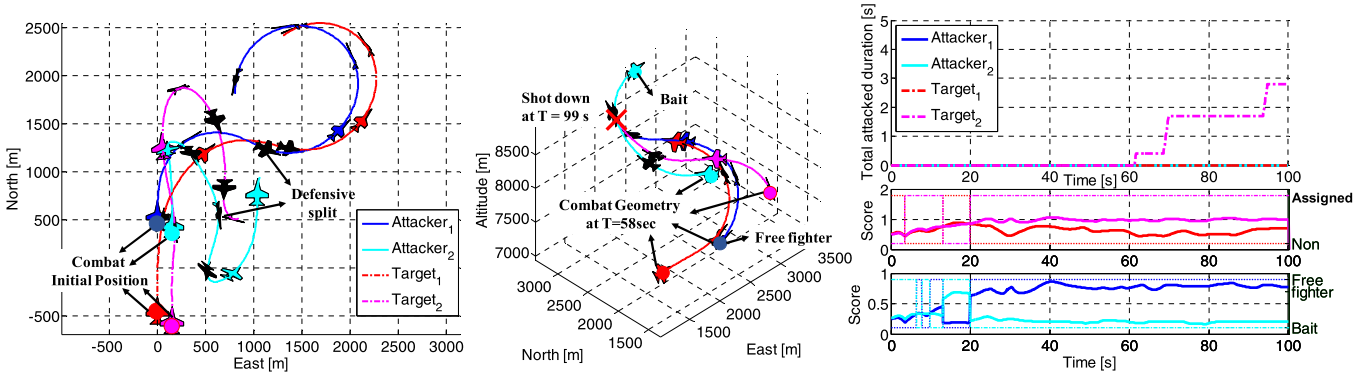
a) Defensive split tactic b) Trajectories from 25 seconds c) Two-on-one combat result plots

Fig. 10. Scenario 2 – Two-on-one tail-chased combat simulation results.



a) Scatter plot of the total attacked duration time b) Bar graph of the simulation results

Fig. 11. Results of the Monte-Carlo simulations for two-on-one combat.



a) Defensive split tactic b) Cooperative tactical maneuver c) Two-on-two combat result plots

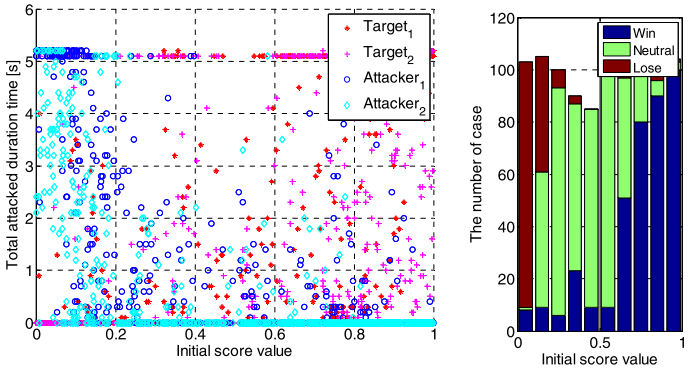
Fig. 12. Scenario 3 – Two-on-two tail-chased simulation results.

Table 2
Required computational times for virtual combat situations.

Simulation scenarios	Required computational time for 1 second virtual combat [s]				Real-time availability
	Model update	Proposed algorithm	Pursuit guidance	Total	
#1. 1:1	0.102	0.066	0.025	0.193	Available
#2. 2:1	0.173	0.133	0.032	0.338	Available
#3. 2:2	0.235	0.135	0.0430	0.413	Available

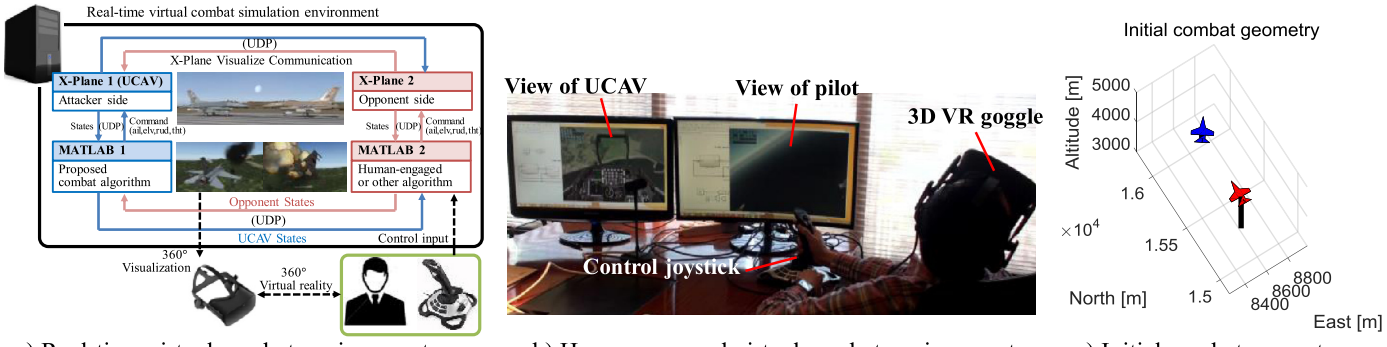
Table 2 summarizes the computational speed for each virtual combat scenario. The proposed algorithm is sufficiently fast compared with pursuit guidance, which is widely used in missile guidance because of its validated calculation speed and high performance. Not only can the algorithm be applied to two-on-one and two-on-two combat situations, but can also be expanded to more complicated engagement environments and can rapidly perform calculations even in scenarios that require complex situation assessment and role assignment.

The autonomous combat algorithm performed not only BFM and energy-management-based maneuvers in the three virtual battle scenarios, but also effective tactical maneuvers with target se-



a) Scatter plot of the total attacked duration time b) Bar graph of the simulation results

Fig. 13. Results of the Monte-Carlo simulations for two-on-two combat.



a) Real-time virtual combat environment b) Human-engaged virtual combat environment c) Initial combat geometry

Fig. 14. Human-engagement simulation environment with VR and virtual combat results.

lection and role assignment in two-on-two environments, thereby exhibiting universal validity for randomly engaged initial conditions, and a light computation load enabling a real-time computation.

4.3. Real-time virtual engagement

A real-time simulation environment with an X-Plane simulator [36] and MATLAB in Fig. 14a) was constructed to validate the practical performance of the proposed algorithm in actual situations. The proposed algorithm was validated by comparing the autonomous combat performance through virtual engagements with another algorithm proposed in [25]. In addition, an Air Force combat instructor of F-16 evaluated the proposed algorithms presented in this paper using a setup based on X-Plane simulator with a VR goggle for realism. The initial combat conditions were set to the beam position commonly used for pilot training, as shown in Fig. 14c), where the opponent located 1000 m above and at a 1400 m distance is chasing the UCAV with a 30° aspect angle.

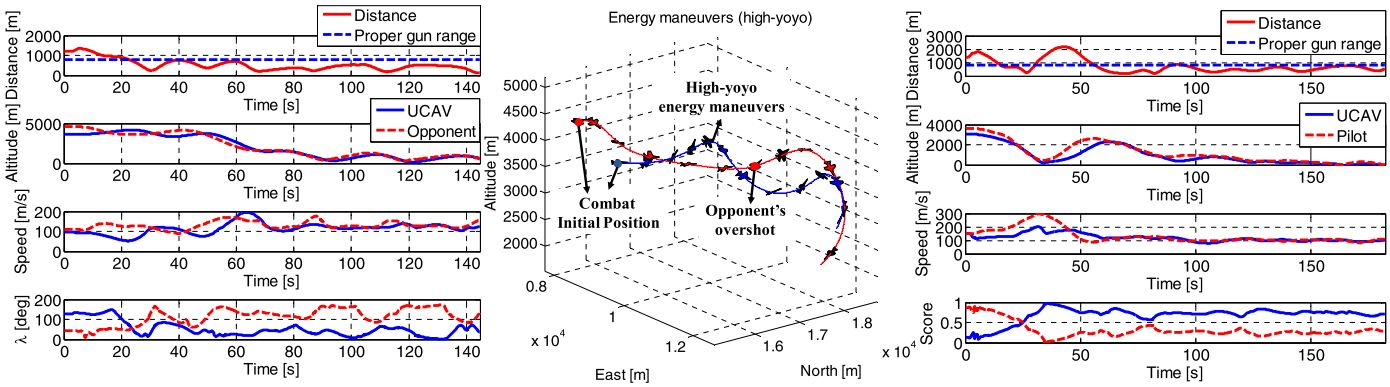
The aggressor piloted by the proposed algorithm successfully reversed the given adverse initial condition of the score value of 0.25, outmaneuvered the opponent, and was eventually shot down at 144 seconds. Especially, UCAV performed impressive maneuvers that reversed the situation by inducing the opponent's overshoots through energy-management as shown in Fig. 15a). In addition, the expert human pilot tested and confirmed that the proposed algorithm performed proper situation assessment and combat maneuvers without mistakes and did not fall for the attempted deceptions of the human pilot, as shown in Fig. 15b).

The Air Force combat instructor in active duty evaluated the simulator and algorithm as follows:

- The research on gun-based close-range engagements is highly relevant.
- The simulation environment provides a realistic 360° combat environment using a VR goggle. It offers a reasonable and practical method for the analysis of the proposed algorithm. This will be helpful in improving training efficiency and the skill of pilots if introduced into the air force.
- Based on experiences as a fighter pilot, the UCAV maneuvers are very similar to the BFMs performed by actual pilots and to be comparable to the pilots who have completed basic combat training.
- Human pilots can easily make mistakes or undesirable maneuvers. Hence, this algorithm can be advantageous in terms of maneuvering according to the results, indicating that it accurately reflects the model features and BFMs.

5. Conclusion

An autonomous combat algorithm for future UCAVs performing gun-based two-on-two aerial combat was proposed in this study. It was designed based on BFM, thus guaranteeing the high-performance of combat maneuvers to supplement previous studies. The proposed algorithm consists of modularized components, each representing the sequential processes of situation assessment, target prediction, and aerial combat guidance. Following situation assessment, a quantified score for the combat situation was obtained through analysis of the combat geometry, and target selection and



a) Combat results and b) energy maneuver against the opponent algorithm in [25] c) Combat results against the human pilot

Fig. 15. Real-time virtual engagement results and trajectories.

role assignment were performed based on this score. In addition, the target prediction module was designed to obtain the future target position based on the target maneuver intentions and maneuvering plane and to generate the impact point for a gun shot through gun modeling. The V–n diagram and E–M chart of the F-16 open model were plotted and applied to the design of the combat guidance law by analyzing the corner velocity, maximum turn rate, and maximum maneuvering point, which are the key factors influencing BFM.

Battles were simulated to validate the energy management performance based on the BFM and the engaging ability of the designed algorithm through three different scenarios: a one-on-one high yo-yo maneuver, a two-on-one numerical superiority situation, and a two-on-two complicated environment. The simulated combat based on a high-fidelity 5-DoF F-16 model confirmed that the designed algorithm can outmaneuver the enemy using BFM and energy management that consider the model characteristics and cooperative tactical maneuvers, thereby showing universal validity with one thousand Monte-Carlo simulations and light computational cost enabling real-time application. The real-time simulation environment was established based on VR and X-Plane. Moreover, a basic or higher-level combat performance was demonstrated through virtual engagements with a comparative algorithm and an Air Force combat instructor.

The designed algorithm can easily be expanded and modified because of its modular structure, and high-performance maneuvering can be performed in real-time based on the BFM. The designed algorithm is expected to be a novel and promising milestone in autonomous aerial combat research.

Conflict of interest statement

The researchers claim no conflicts of interest.

Acknowledgements

This work was supported by the Agency for Defense Development under contract UD150047JD. We would like to thank the participants for conducting the interactive virtual combats with the proposed algorithm and for the evaluation based on the knowledge of an expert.

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